

Sales CSV Cleaner & Validator

Problem

Sales CSV files often contain real-world messiness:

- Duplicates
- Missing values
- Inconsistent column names and formatting
- Mixed date formats
- Numeric values with commas or invalid entries

Without cleaning, analysis or automation workflows can fail or give wrong results.

Solution

A Python script using Pandas that:

- Detects and removes duplicate rows
- Fills missing values in text and numeric columns
- Standardizes headers (strip spaces, lowercase, remove special characters)
- Normalizes date columns and fills missing dates
- Converts numeric columns to proper types, handles commas/invalid values
- Clips negative numbers
- Recalculates Total Revenue column if needed
- Produces a cleaned CSV ready for analysis

Results

Applied on a sample CSV, the script:

- Removed duplicate rows
- Filled missing values with appropriate defaults
- Standardized headers and text columns
- Converted dates and numeric columns correctly
- Recalculated Total Revenue
- Saved cleaned CSV for analysis

Usage

1. Place your CSV in the data/ folder
2. Run the script:

```
python scripts/csv_cleaner.py
```
3. The cleaned CSV will be saved in data/sales_data_clean.csv